

Zero Trust on Z: Identity and Governance

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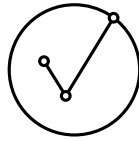


Security challenges in IBM Z environments



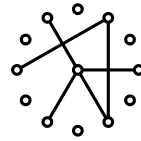
Threat Landscape

The threat landscape has evolved. While IBM Z® offers robust defenses. Clients must take deliberate steps to secure their environments



Compliance

As compliance requirements evolve to include mainframe environments, meeting these requirements demands a more integrated and proactive approach



Complexity

Increasingly hybrid environments that include mainframe applications create challenges in visibility and control across a client's infrastructure¹



Skills

As specialized skills become increasingly scarce, clients need to adopt new strategies and technologies to ensure continued protection of their environments

2

The Cybersecurity Framework context



Identify and understand security risks in your environment



What

- Discover the systems and data that require security protection
- Identify potential vulnerabilities and threats
- Establish the baseline for other cybersecurity actions

How

- Data inventory and classification
- Security and compliance assessments
- Penetration testing

Benefits

- Creates a targeted security strategy to protect sensitive assets
- Prioritizes risk mitigation efforts and resource allocation
- Stay compliant to minimize legal and financial risk

Compliance

- Maps assets and risks to regulatory needs
- Defines roles and policies for audit readiness
- Aligns security with business and compliance goals

Identify solutions and services

Data Discovery & Inventory

[IBM Z Crypto Discovery & Inventory](#)

Simplifies how clients discover cryptographic assets on the Z platform and prepares them for their **post-quantum crypto journey**

Compliance simplification

[IBM zSecure Compliance](#)

Collects compliance data from z/OS subsystems to simplify audits, **saving time and effort**. Locates sensitive data, potential vulnerabilities & other compliance issues using automated reports.

Security assessments & penetration testing

Technical Expert Labs

Provides multiple levels of security assessments in the context of z/OS. New penetration testing services are being explored as well.

The Cybersecurity Framework context



Protect your data and risk exposure by putting safeguards in place



What

- Ensure authorized access to data
- Reduce risk exposure to cyber threats
- Defend against “Harvest now, decrypt later” attacks with Quantum Computing

Benefits

- Reduce the impact and cost of data breaches
- Protect brand reputation and maintain client trust
- Stay compliant to minimize legal and financial risk

How

- Identity and access management
- Encryption & Secrets Management
- Data Protection

Compliance

- Implements safeguards to control access and protect sensitive data.
- Enforces policies for secure behavior and awareness training.
- Applies encryption and backups to ensure data confidentiality and availability

Protect solutions

Identity and Access Management

[IBM Z Multi-factor Authentication](#)

Mitigate the risk of compromised passwords with expanded authentication capabilities for your mission-critical IBM Z environments

[IBM zSecure® Admin](#)

Automates IT security management tasks and allows administering multiple systems from a single interface. It helps identifying, analyzing and preventing problems in RACF®.

Encryption & Secrets Management

[IBM Unified Key Orchestrator](#)

Centrally orchestrates and secures the lifecycle of encryption keys across your enterprise

[IBM Vault Self-Managed](#)

Identity-based secrets management system to centrally store and manage all your business secrets. Leverage it on z/OS with the **IBM zSecure Secret Manager**.

Data Protection

[IBM Guardium® Data Protection](#)

Guards data stores in a hybrid cloud through capabilities like data activity monitoring and analytics, and automated compliance auditing and reporting.

Strengthen authentication –on both z/OS & Linux on Z

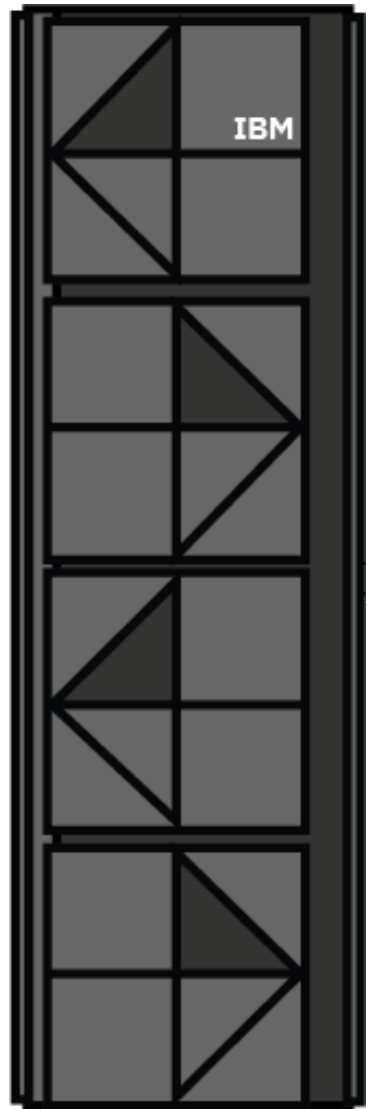
Advocated by Zero Trust principles, ransomware mitigations, & rising regulations...



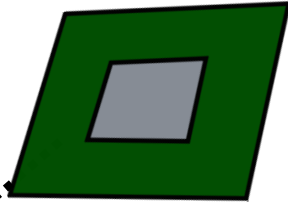
MFA

8-character password authentication constitutes significant risk.

Post-quantum crypto encryption components

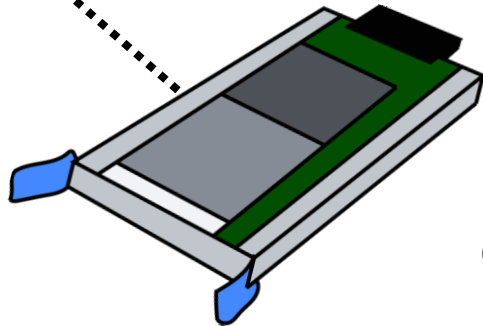


CPACF

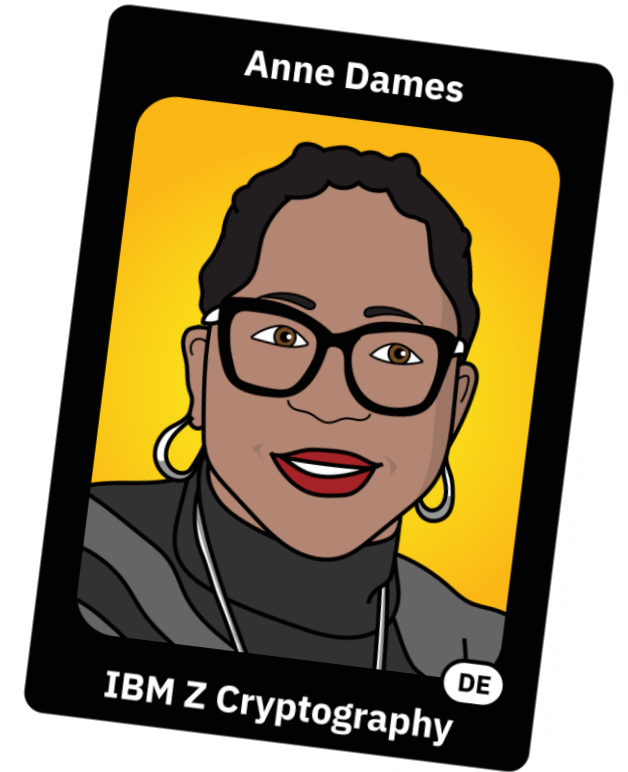


High performance
key calculations

Crypto Express 8S (CEX8S)



High security key
calculations

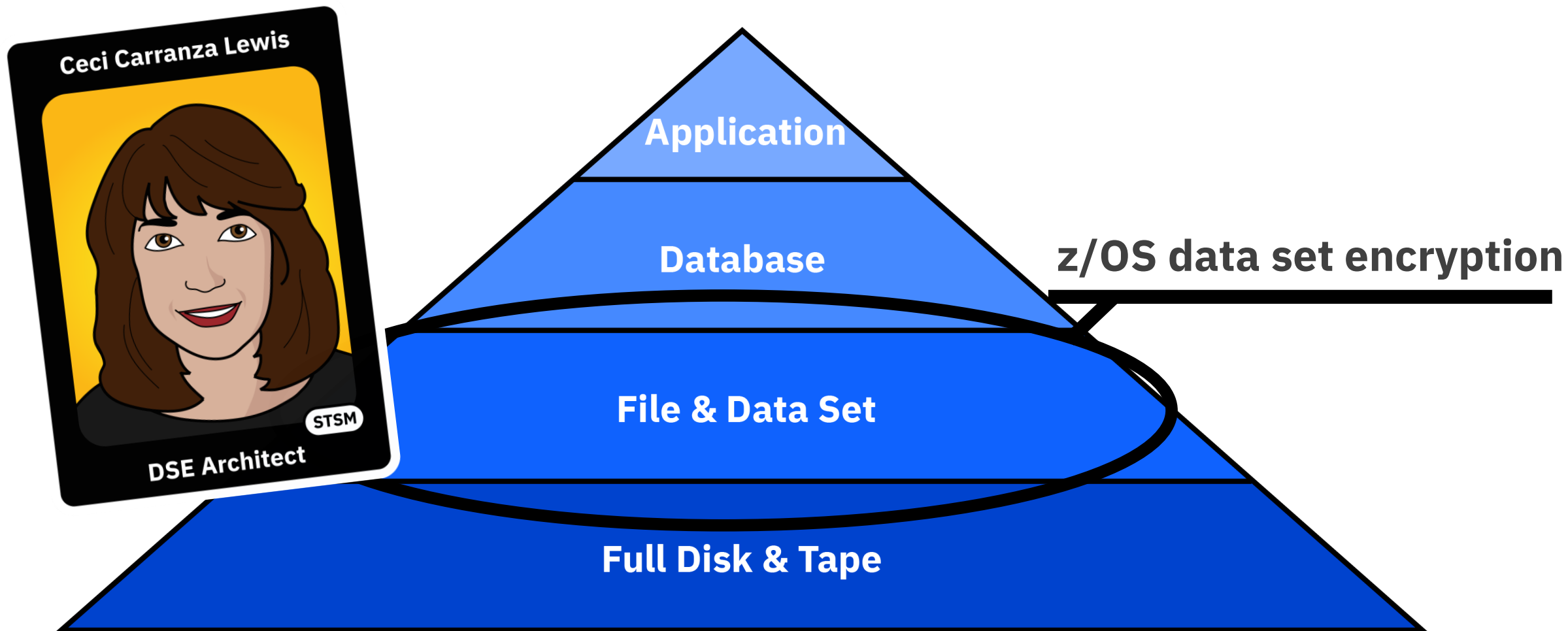


TKE Workstation



Simplified &
secure Master
Key usage

Encryption Pyramid



RACF Statement of Direction realized

Encrypted VSAM data set support in RACF

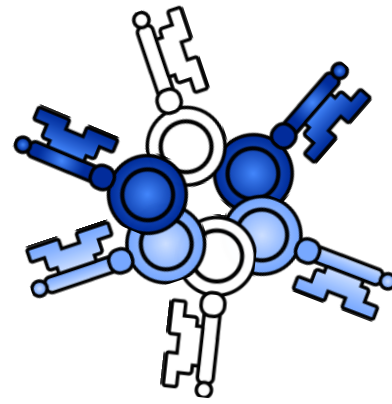
“IBM intends to enhance pervasive encryption through RACF support for the use of an encrypted VSAM data set as its data base in specific configurations.”

Why VSAM?

- Enables data set encryption
- Integrates well with RACF’s existing serialization
- Consistent with RACF’s current database architecture
- Provides the ability to utilize existing diagnostics
- Leverages standard z/OS skills
- Leverages current and future I/O infrastructure improvements



Key Management



Unified Key Orchestrator for IBM z/OS (was EKMF Web)
Operational Keys

Trusted Key Entry (TKE) Workstation
Master Keys

Guardium Key Lifecycle Manager
(GKLM)
Self-encrypting Device Keys

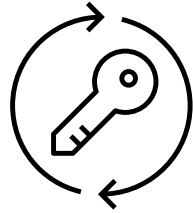
There's another form of secret

That plays a pivotal role for post-quantum crypto

That can also provide authentication for a machine

In need of its own management simplification...

The matter of certificates...



Maximum lifetime of a public TLS certificate...

Previously:	398 days
March 15, 2026:	200 days
March 15, 2027:	100 days
March 15, 2029:	47 days

How many certificates do you have in your RACF database?

How many certificates do you NOT have in your RACF database?

Are you ready?

~10x

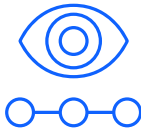
Increase in manual
certificate renewal costs
expected by 2029
without automation.**

*See its announcement with statement of direction details [here](#)

**Based on customers polled, who expect to renew certificates monthly

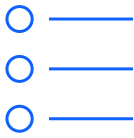
IBM zSecure Secret Manager

Vision*



Policy-Driven Certificate Auto-Renewal

Administrators define renewal policies directly within IBM zSecure Secret Manager from its initial release. Certificate expiration is then continuously monitored and automatically renewed based on that policy, leveraging IBM Vault Self-Managed for Z and LinuxONE as the internal Certificate Authority (CA).



Automated Certificate Deployment*

IBM zSecure Secret Manager plans to expand its scope to leverage external Certificate Authorities (CA's) to include public certificates within z/OS, and with that it also plans to extend simplification measures to help automate certificate deployment.



Best Kept Secrets*

IBM zSecure Secret Manager plans to further bridge z/OS and the RACF security database to the enterprise-level operations of IBM Vault Self-Managed for Z and LinuxONE with the management of z/OS-custom secrets, providing greater security and efficiency.

IBM Vault Self-Managed for IBM Z & LinuxONE

IBM Vault Self-Managed for Z and LinuxONE is built to protect your enterprise secrets, utilizing the industry's most reputable platform for performance, availability, reliability and security.

Why IBM Vault Self-Managed for Secrets Management?

- ✓ **Identity-based security** uses trusted identities to automate access to secrets, data and applications
- ✓ **Security system of record** to centrally store and protect secrets across clouds and applications.
- ✓ **Lifecycle management** of one's credentials to ensure proper oversight, rotation and expiry

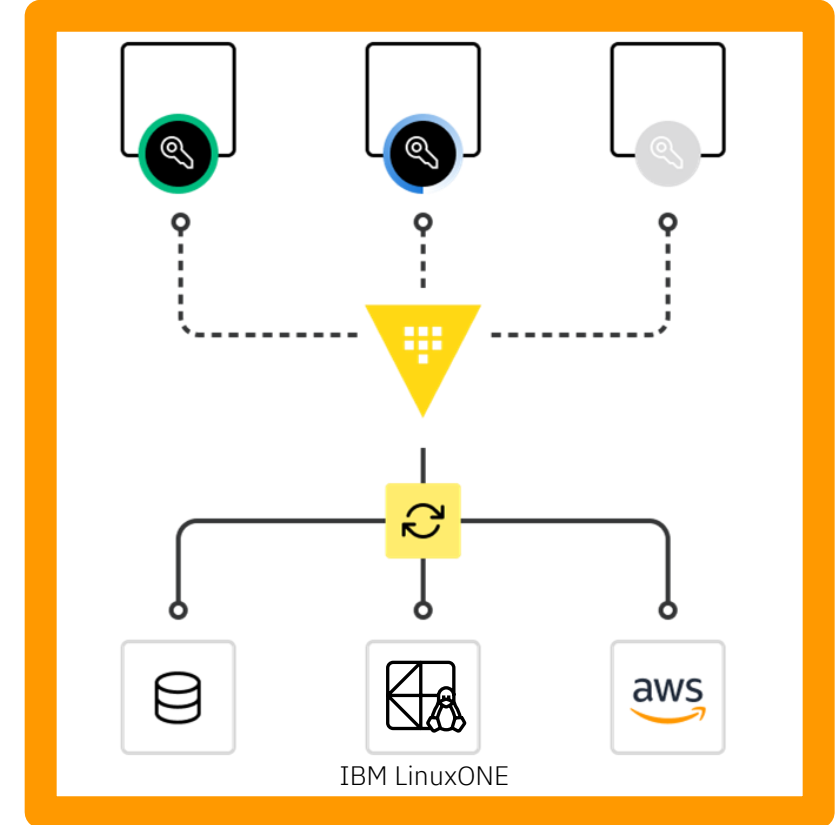


Figure 1. Centrally manage and enforce access to secrets and systems based on trusted sources of application and user identity.

IBM Vault Self-Managed on IBM Z and LinuxONE

Extending IBM Z & LinuxONE unique capabilities to Vault

Confidential Computing Environment

- What: Organizations can run Vault within a Trusted Execution Environment (IBM Secure Execution)
- Outcome: Ensuring that the *Vault process and secrets stored are encrypted in memory and protected* from unauthorized access

Encrypted Communication

- What: Vault can be configured to operate within an encrypted enclave, where only authenticated workloads running inside a TEE can retrieve secrets
- Outcome: *Secrets are never exposed in plaintext* outside the enclave

IBM Crypto Express, Hardware Security Modules (HSMs)

- What: Leverage HSM-backed Vault instances
- Outcomes:
 - Store and manage cryptographic keys *securely*
 - Ensure that key material is *never exposed*, even to Vault administrators
 - *Secured, automatic* sealing and unsealing of secrets engines

Additional Benefits

IBM LinuxONE industry leading Resiliency

- What: IBM LinuxONE architecture designed to deliver 99.999999% availability, ensuring continuous operation
- Outcome: *Enhances Vaults resiliency and downtime*

Certificate Management

As-Is Scenario

Identify certificates to be renewed

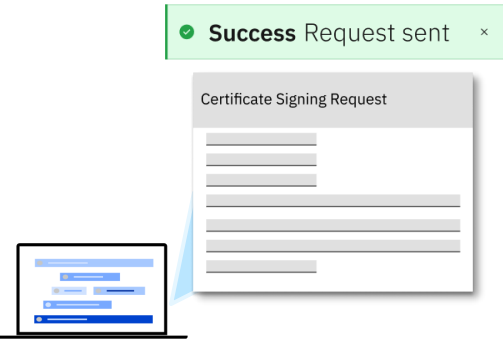


The Application Owner is responsible for tracking the certificates lifecycle and driving renewal.



Conrad
Application Owner

Begin the certificate renewal process



The Application Owner, enters the required certificate information to create the CSR or Certificate Signing Request to obtain a digital certificate.



Conrad
Application Owner

Renew certificate with CA
(Certificate Authority)



The Certificate Authority, receives and reviews the CSR request.



Conrad
Application Owner

Approves the CSR
(Certificate Signing Request)



Issues the signed certificate and sends to requester.



Conrad
Application Owner

Certificate Management

As-Is Scenario

Receives the certificate

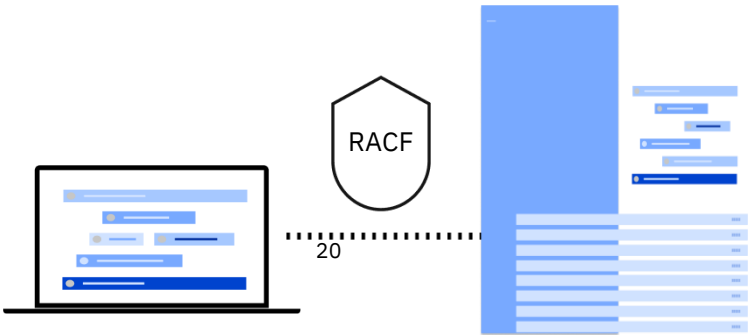


The Application Owner, receives the certificate from the CA and reviews the certificate information.



Conrad
Application Owner

Install the certificate



Install the certificate in the RACF, JAVA, or UNIX certificate stores.



Conrad
Application Owner

Refresh application



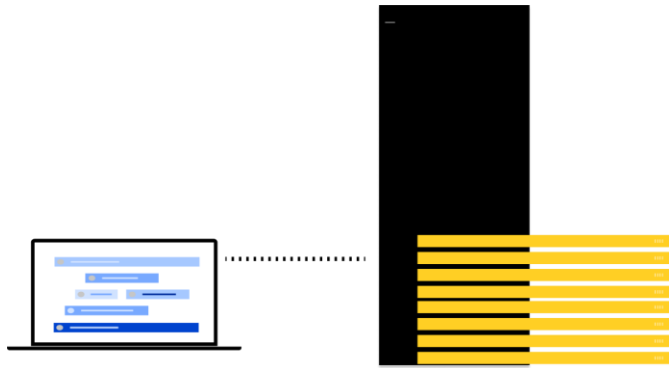
Configure application with the certificate key store information. Refresh application view of the key ring.



Conrad
Application Owner

TO- BE Certificate Management

Configure auto-renew environment



The Security Admin will set up IBM zSecure Secret Manager to issue certificates with IBM Vault Self-Managed for Z and LinuxONE.



Maya
Security Admin

- Establish IBM Vault Self-Managed for Z and LinuxONE as a PKI
- Configure zSecure Secret Manager and connect to IBM Vault Self-Managed for Z and LinuxONE

Create auto-renew policy for each certificate



The Application Owner will set the new auto-renew policy using the new security interface.



Conrad
Application Owner

IBM zSecure Secret Manager will:

- Register the certificate for auto-renewal
- Monitor for certificate expiration
- Renew certificate automatically
- Issue notification (message, SMF) for applications to refresh

Refresh / Restart application with certificate



Applications that auto-refresh*:

- No manual intervention for certificate renewal

Applications that don't auto-refresh:

- No manual steps are required for the certificate renewal process, just the application refresh
- Downtime is minimal

Thank you

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